

ECEN 743: Reinforcement Learning

Spring 2025

Course Information

Course Number:	ECEN 743
Course Title:	Reinforcement Learning
Section:	600
Time:	MW 05:45PM - 07:00PM
Location:	ETB 1037
Credit Hours:	3 Credit Hours
Learning Management System:	Canvas

Instructor Details

Instructor:	Dr. Dileep Kalathil
Office:	334G WEB
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Office Hours:	MW 4:30PM - 5:30PM
Office Hours Location:	334G WEB

Course Description

Introduction to the theory and practice of reinforcement learning; Topics include Markov decision processes; dynamic programming; Q-Learning; policy gradient algorithms; deep reinforcement learning; imitation learning; meta learning; multi-agent learning.

Course Prerequisite

Graduate classification. Familiarity with the undergraduate level probability theory. Programming skills in a high-level language. Programming assignments will be in Python. Familiarity with neural networks training.

Detailed Course Description

Engineering systems are moving towards an autonomous future. One key aspect of any autonomous system is the intelligent agents who can adaptively learn and take sequential decisions in an uncertain and stochastic environment. Reinforcement learning (RL) is a powerful paradigm for modeling such intelligent systems and to learn optimal control policies. RL is extremely useful in a number of real-world applications, starting from the classical stochastic control problems to most recent applications in robotics, large language models, healthcare, and autonomous vehicles. This course offers a rigorous introduction to the theory and practice of RL.

Course Learning Outcomes

After the completion of this course, students will be able to:

1. Model a stochastic control problem as an MDP and solve using dynamic programming
2. Analyze the performance of standard RL algorithms
3. Implement deep neural network-based RL algorithms
4. Develop advanced RL algorithms, including offline RL, RLHF, and meta-RL

Textbook and/or Resource Material

This course has no *required* textbooks. The instructor will use materials from multiple textbooks, papers, and other resources. However, students are strongly encouraged to refer/read one or more of the following textbooks

1. [SB] *Reinforcement Learning: An Introduction*, Richard S. Sutton and Andrew G. Barto, MIT Press, Cambridge, MA, 2018. PDF version available [here](#)
2. [BRL] *A Course on Reinforcement Learning*, Dimitri Bertsekas, Athena Scientific, 2024. PDF version available [here](#)
3. [KM] *Reinforcement Learning: An Overview*, Kevin Murphy, 2024. PDF version available [here](#)
4. [BDP] *Dynamic Programming and Optimal Control*, Vol. 2, Dimitri Bertsekas, Athena Scientific, 2012.
5. [MMT] *Reinforcement Learning: Foundations*, Shie Mannor, Yishay Mansour, and Aviv Tamar. PDF version available [here](#).
6. [AJKS] *Reinforcement Learning: Theory and Algorithms*, Alekh Agarwal, Nan Jiang, Sham M. Kakade, Wen Sun. Working draft. PDF version available [here](#).

Grading Policy

- Assignments: 35%
- Exam 2: 15%
- Exam 1: 15%
- Class Project: 35%

Class project points are divided into:

- Project proposal: 5%
- Final project presentation and report: 20%
- Project milestone presentation: 10%

If your overall grade falls within one of the prescribed ranges, then you are guaranteed to receive at least the letter grade indicated.

A: 90-100%

B: 80-89%

C: 70-79 %

D: 60-69 %

F: 0-59 %

The Academic Rules website at Texas A&M University, and its section on Grading in particular, discusses possible grades and their respective meaning: <https://student-rules.tamu.edu/rule10>.

Exam Schedule:

- Exam 1: February 19
- Exam 2: March 26
- Project proposal: March 31
- Project milestone presentation: April 14
- Final project presentation and report: May 1

Late Work Policy

Assignments turned in late will not be accepted. It is your responsibility to ensure that you are available and present for the exams. Exceptions will be made to the rules above only in order to accommodate special circumstances.

Course Schedule

Week	Topics
1	Introduction to RL Essential Markov Chains for RL (Reading) Introduction to Markov Decision Process
2	Policy Evaluation Fixed Point and Contraction Mapping
3	Value Iteration and Policy Iteration
4	Tabular Temporal Difference (TD) Learning Tabular Q-Learning
5	RL with Function Approximation
6	Deep Q-Learning
7	Policy Gradient Algorithms
8	Advanced Policy Gradient Algorithms
9	Exploration and Multi-Armed Bandits
10	Monte-Carlo Tree Search AlphaGo and variants
11	Imitation Learning Inverse RL
12	Offline RL
13	Reinforcement Learning with Human Feedback
14	Meta-learning
15	Final project presentation and report

University Policies

This section outlines the university level policies that must be included in each course syllabus. The TAMU Faculty Senate established the wording of these policies.

Attendance Policy

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments.

Please refer to Student Rule 7 in its entirety for information about excused absences, including definitions, and related documentation and timelines.

Makeup Work Policy

Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student's grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor.

Please refer to Student Rule 7 in its entirety for information about makeup work, including definitions, and related documentation and timelines.

Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor (Student Rule 7, Section 7.4.1).

"The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence" (Student Rule 7, Section 7.4.2).

Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See Student Rule 24.).

Academic Integrity Statement and Policy

"An Aggie does not lie, cheat or steal, or tolerate those who do." <http://aggiehonor.tamu.edu>.

"Texas A&M University students are responsible for authenticating all work submitted to an instructor. If asked, students must be able to produce proof that the item submitted is indeed the work of that student. Students must keep appropriate records at all times. The inability to authenticate one's work, should the instructor request it, may be sufficient grounds to initiate an academic misconduct case" Section 20.1.2.3, Student Rule 20).

You can learn more about the Aggie Honor System Office Rules and Procedures, academic integrity, and your rights and responsibilities at <http://aggiehonor.tamu.edu>.

Americans with Disabilities Act (ADA) Policy

Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact the Disability Resources office on your campus (resources listed below) Disabilities may include, but are not limited to attentional, learning, mental health, sensory, physical, or chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.

Disability Resources is located in the Student Services Building or at (979) 845-1637 or visit <http://disability.tamu.edu>.

Title IX and Statement on Limits to Confidentiality

Texas A&M University is committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws prohibit gender-based discrimination and sexual harassment, including sexual assault, sexual exploitation, domestic violence, dating violence, and stalking.

With the exception of some medical and mental health providers, all university employees (including full and part-time faculty, staff, paid graduate assistants, student workers, etc.) are Mandatory Reporters and must report to the Title IX Office if the employee experiences, observes, or becomes aware of an incident that meets the following conditions (see University Rule 08.01.01.M1)

- The incident is reasonably believed to be discrimination or harassment.
- The incident is alleged to have been committed by or against a person who, at the time of the incident, was (1) a student enrolled at the University or (2) an employee of the University.

Mandatory Reporters must file a report regardless of how the information comes to their attention – including but not limited to face-to-face conversations, a written class assignment or paper, class discussion, email, text, or social media post. Although Mandatory Reporters must file a report, in most instances, you will be able to control how the report is handled, including whether or not to pursue a formal investigation. The University’s goal is to make sure you are aware of the range of options available to you and to ensure access to the resources you need.

Students wishing to discuss concerns in a confidential setting are encouraged to make an appointment with Counseling and Psychological Services (CAPS).

Students can learn more about filing a report, accessing supportive resources, and navigating the Title IX investigation and resolution process on the University’s Title IX webpage.

Statement on Mental Health and Wellness

Texas A&M University recognizes that mental health and wellness are critical factors that influence a student’s academic success and overall wellbeing. Students are encouraged to engage in healthy self-care practices by utilizing the resources and services available through University Health Services. The TELUS Health Student Support app provides access to professional counseling in multiple languages anytime, anywhere by phone or chat, and the 988 Suicide & Crisis Lifeline offers 24-hour emergency support at 988 or 988lifeline.org

Statement on the Family Educational Rights and Privacy Act (FERPA)

FERPA is a federal law designed to protect the privacy of educational records by limiting access to these records, to establish the right of students to inspect and review their educational records and to provide guidelines for the correction of inaccurate and misleading data through informal and formal hearings. Currently enrolled students wishing to withhold any or all directory information items may do so by going to <https://howdy.tamu.edu/> and clicking on the “Directory Hold Information” link in the Student Records channel on the MyRecord tab. The complete FERPA Notice to Students and the student records policy is available on the Office of the Registrar webpage.

Items that can never be identified as public information are a student’s social security number,

citizenship, gender, grades, GPR or class schedule. All efforts will be made in this class to protect your privacy and to ensure confidential treatment of information associated with or generated by your participation in the class.

Directory items include name, UIN, local address, permanent address, email address, local telephone number, permanent telephone number, dates of attendance, program of study (college, major, campus), classification, previous institutions attended, degrees honors and awards received, participation in officially recognized activities and sports, medical residence location and medical residence specialization.